

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (Civil) (Credit System) Degree Examinations**

May - June 2023

20CV604 - ENVIRONMENTAL ENGINEERING - II

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I**Marks BT* CO* PO***

1. a) Explain dry weather flow and storm water flow. List the factors affecting dry weather flow and storm water flow.
- b) Water was supplied at a per capita rate 135 LPCD having total population of 40000 persons to a newly developed township. A sewer must be designed to be running full at maximum discharge. Compute the diameter of the sewer and estimate the velocity of flow when running full. Take $n = 0.013$ at all flow depth, slope of 1 in 625. Assume 80% of water supplied appears as sewage flow, design flow = 4 times the dry weather flow and sewer running full.

10 L*2 1 1

10 L3 2 3

2. a) A catchment area of a town on which rain falls is classified as follows:

S. No.	Type	% Area	Runoff Coefficient
1	Roof	20	0.9
2	Pavement	30	0.8
3	Road	30	0.4
4	Vacant Plot	20	0.1

Take, Rainfall intensity = 40 mm/hr; Total area = 200 ha
 Domestic sewage = 80% of water supplied and a peak factor of 2.0.
 If population density = 250/ha and maximum rate of supply of water = 150 LPCD, compute the analysis of sewage quantity for

i) Separate System

ii) Partially Separate System

10 L3 1 2

- b) With a neat sketch of hydrograph, explain time of concentration.

10 L2 2 1

3. a) Estimation of sewage must be done for a city having population density of 350 persons/ha. Water supply is at the rate of 150 LPCD and the peak factor 1.5. The catchment area of city is 300 ha with the following classification.

Surface	% Area	Runoff coefficient
Roofs	20	0.95
Pavement	15	0.80
Garden	30	0.20
Roads	20	0.40
Plots	15	0.15

If the sewage flow = 0.8 times the water supplied, calculate the average impervious factor and total quantity of sewage considering the rainfall of 40 mm/hr. Comment on the quantity of storm water with respect to the sanitary sewage.

10 L3 1 2

- b) Discuss the importance of self-cleansing and non-scouring velocities in sewers.

10 L2 2 1

Unit – II

4. a) Describe the various forces which affect the self-purification process.

10 L2 3 1

b) The sewage flow from a township is 6MLD. Calculate the diameter of a primary sedimentation tank with a detention time of 120 minutes and overflow rate of 1500 L/m ² /hr. Estimate the weir loading for the tank designed and compare it with the permissible limits.		10	L3	4	8
5.	a) Illustrate the formulation of I-stage BOD Curve using a hypothetical BOD reaction curve. List the applications of BOD data.	10	L2	3	1
	b) A rectangular grit chamber is designed to remove particles with a diameter 0.2mm and specific gravity 2.65. The settling velocities of these particles are found to be 0.02m/s. A flow through velocity of 0.3m/s will be maintained by the proportioning weir. Determine the channel dimensions for a maximum wastewater flow of 10000m ³ /d.	10	L4	4	3
6.	a) Explain sewage sickness and list its preventive measures.	10	L2	3	6
	b) Design a suitable dimension of circular trickling filter units for treating 5MLD sewage. The BOD of sewage is 150 mg/L. Take organic loading rate = 1500 kg/ha_m/d	10	L3	4	3
Unit – III					
7.	a) With a neat sketch, explain the construction and process involved in a septic tank.	10	L2	5	6
	b) Discuss briefly about the recycle and reuse of water for various applications.	10	L6	5	4,5
8.	a) Illustrate the mechanism of anaerobic digestion process using a flow chart and mention its optimum conditions.	10	L2	5	1
	b) With the help of neat sketches, discuss the various methods of modifications of activated sludge process.	10	L4	5	1,5

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
